

1.		5
1.1.		5
1.2.		5
2.		6
2.1.		6
2.2.		6
2.3.		6
2.4.		7
2.5.		7
2.6.		7
2.7.		7
2.8.		8
3.		9
3.1.		9
3.2.		9
3.3.		9
3.4.		9
3.5.		10
3.6.		10
3.7.		10
3.8.		10
3.9.		11
3.10.		11
3.11.		11
3.12.		11
3.13.		12

3.14.		12
3.15.		12
3.16.		13
3.17.		13
3.18.		13
3.19.		14
3.20.		14
3.21.		15
3.22.		15
3.23. n		15
4.		16
4.1.		16
4.2.		16
4.3.		16
4.4.		16
4.5.		17
4.6.		17
4.7.		17
4.8.		18
4.9.		18
4.10.		18
4.11.		19
5.		20
5.1.		20
5.2.		20
5.3.		20
5.4.		20
5.5.		21
5.6.		21
5.7.		21
5.8.		22

6.		23
6.1.		23
6.2.		23
6.3.		23
6.4.		23
6.5.		23
6.6.		24
6.7.		25
6.8.		25
6.9.		25
6.10.		26
6.11.		26
6.12.		27
6.13.		27
6.14.		27
6.15.		27
7.		29
7.1.		29
7.2.		29
7.3.		30
7.4.		30
7.5.		30
7.6.		30
7.7.		31
7.8.		31
7.9.		31
7.10.		31
7.11.		32
7.12.		32
7.13.	2	32
7.14.		33

7.15.		33
-------	-------	----

1.

1.1.

2022

pp. 40 - 41

1.2.

2022

pp. 43, 47 - 54

12

$X \text{ cm}^2$

$Y \text{ g}$

(1) $Y = aX + b$

(2)

(3)

$$(X, Y) = (11, 23), (23, 49), (5, 10), (50, 115), (32, 64), (21, 45) \\ (16, 35), (39, 84), (25, 50), (45, 100), (5, 11), (34, 70)$$

2.

2.1.

2022

2022

2022

p. 65

$$\binom{1}{1}$$

$$\binom{1}{1}$$

$$\binom{2}{2}$$

$$\binom{2}{2}$$

$$\binom{2}{2}$$

$$\binom{1}{1}$$

$$\binom{2}{2}$$

$$\binom{3}{3} = 1$$

$$6 \cdot \binom{2}{2}$$

$$1 \ 2 \ 3 \ 4 \ 5 \ 6$$

$$\binom{2}{2}$$

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\binom{3}{3}$$

$$\binom{2}{2} = \Omega$$

$$\binom{1}{1}$$

$$\binom{3}{3}$$

$$\binom{2}{2}$$

2.2.

2022

2022

2022

pp. 78, 80, 83

$$(1)$$

$$(2)$$

$$(3)$$

$$(4) \ 3$$

2.3.

2022

2024

2026

p. 78

$$1$$

$$X$$

$$c$$

$$P(X = x) = c \times x \quad (x = 1, 2, 3, 4, 5, 6) \quad \times$$

$$(1) \quad c$$

$$(2) \quad x = 1, 2, 3, 4, 5, 6 \quad P(X = x)$$

$$(3) \quad X \quad E[X]$$

$$(4) \quad P(X \geq 4)$$

2.4.

2023

pp. 81, 86

$$f(x) = \begin{cases} \frac{c}{x^3} & (x \geq 10) \\ 0 & (x < 10) \end{cases}$$

$$(1) \quad c$$

$$(2) \quad F(x)$$

$$(3) \quad P(x \geq a) = \frac{1}{2} \quad a$$

2.5.

2024

pp. 81, 86

 c

$$f(x) = \begin{cases} c(1-x) & (0 \leq x \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(1) \quad c$$

$$(2) \quad P\left(\frac{1}{2} < x \leq \frac{2}{3}\right)$$

$$(3) \quad F(x)$$

$$(4) \quad P\left(\frac{1}{2} \leq x < 3\right)$$

2.6.

2022

2023

2024

pp. 134, 151, 180

$$(1)$$

$$(2)$$

$$(3)$$

$$(4)$$

2.7.

2023

2023

pp. 134, 151, 180

$$X_1, X_2, \dots, X_n$$

 Y

2

$$(1) \quad X_1, X_2, \dots, X_n$$

 Y

$$(2) \quad 2 \quad /$$

$$(3)$$

(4)

(5)

(6)

1

2.8.

2025

(1) X μ σ^2

(2) 200 78 60 6 42

3.

3.1.

2024

p. 107

6

4

1

2

(1) 2

(2) 2

3.2.

2026

p. 107

3

5

(1)

1

1

(a) 2

(b) 2

(c) 2

1

(2)

2

1

(3)

(1)

(2)

3.3.

2022

pp. 107 - 108

1 2 3 4

2

$X_1 \quad X_2 \quad Y = |X_1 - X_2|$

(1) Y

(2) Y

(3) Y

3.4.

2023

2025

pp. 113 - 114

(X, Y)

$$f(x, y) = \begin{cases} 6e^{-2x-3y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$X \quad Y$

3.5.

2026

$X \quad Y$

$$\int_0^\infty e^{-x} dx = 1, \quad \int_0^\infty x e^{-x} dx = 1$$

$$(1) f(x, y) = \begin{cases} x e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(a) \quad f_X(x) \quad f_Y(y)$$

$$(b) \quad X \quad Y$$

$$(2) f(x, y) = \begin{cases} c(x+y)e^{-x-y} & (x \geq 0, y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$(a) \quad c$$

$$(b) \quad X \quad Y$$

3.6.

2022

pp. 103 - 104, 113

1 4 2 2 1

$X = 0$ $X = 1$ 2

$Y = 0$ $Y = 1$

$$(1) \quad (X, Y)$$

$$(2) \quad X \quad Y$$

3.7.

2022

pp. 67, 114 - 115

2 $A \quad B$

3.8.

2022

2023

2026

p. 115

3

$\frac{1}{2}$

(1)

1

2

(2)

1

2

A

2

3

B

A

B

(3)

$P(A|B)$

3.9.

2024

p. 115

2

A

4

B

2

7

C

2

9

(1)

A

B

(2)

A

C

3.10.

2023

pp. 72 - 73, 109 - 110, 114 - 115

2

E

F

$P(E) = a$

$P(F) = b$

(1)

$P(E \cup F)$

a

b

(2)

$P(E \cap F^c | E \cup F)$

a

b

F^c

F

3.11.

2025

1

5

X

Y

$Z = X - Y$

(1)

X

Y

(2)

Z

(3)

Z

(4)

Z

3.12.

2025

pp. 118 - 119

(1) 2 A B

(2) 500 1 X X
 2 K_1 K_2 $3 : 1$
 K_1, K_2 0.98 0.90
 0.01
 K_1

3.13.

2022 2023 pp. 118 - 119
 3 5 1
 2
(1) 1
(2) 2 1

3.14.

2023 pp. 118 - 119
 1 10% 3
(1)
(2) 2

3.15.

2022 pp. 118 - 119
 5 4 3
 1 100 g 45 g 30 g
(1) 1
 40 g

(2) 2
100 g 2

(3) 3
150 g 3

3.16.

2022 2024 pp. 118 - 119
A 0.1%
A /
99% 10%
(1)
(2) A 1

3.17.

2023 pp. 118 - 119
99%
99% 0.5%
1

(1)

(2)

(3)

(4)

3.18.

2023 pp. 118 - 119
S S
K

- S K 99% 1%
- S K 99% 1%

(1) K S

(2) (1)

3.19.

2025

pp. 118 - 119

A

B

(1) $P(A|B) = 0.98$ $P(A|B^c) = 0.005$ $P(B) = 0.001$

(2) $P(A|B^c) = 0.003$ $P(B) = 0.0005$ $P(B|A) \geq 0.142$ $P(A|B)$

3.20.

pp. 118 - 119

500 1 XXX 2

K_1 K_2 $K_1 : K_2 = 3 : 1$

K_1 0.98

K_2 0.90

0.01

(1) K_1

(2) (1)

3.21.

2023

pp. 122 - 124

p

$0 < p < 1$

X

Y

$f(k) = \begin{cases} p & (k = 1) \\ 1 - p & (k = -1) \end{cases}$

$Z = XY$

(1)

$E[Z]$

(2)

X

Z

$E[(X - E[X])(Z - E[Z])]$

(3)

X

Z

p

p

Y

Z

(4)

(3)

p

$\Pr[X + Y + Z \leq 2]$

3.22.

2023

2024

pp. 125 - 126

2

X

Y

(1)

a

(2)

X

Y

$\text{Cov}(X, Y)$

(3)

X

Y

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

3.23. n

2023

p. 127

n

1

n

$(n + 1)$

4.

4.1.

2025

2

$B(2, p)$

4.2.

2024

p. 141

1

6

(1)

10

5

(2)

100

25

4.3.

2023

2026

pp. 93 - 95, 141

p

n

X

2

$\text{Bin}(n, p)$

$q = \left(\frac{X}{n}\right)$

(1)

$P(|q - p| \leq 0.02) \geq 0.95$

n

(2)

(1)

n

(2)

(3)

p

(3)

(1)

-

n

(4)

(2)

(3)

n

4.4.

2024

pp. 143 - 145, 153

(1)

:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$y = f(x)$

(x, y)

(2) /

(3)

4.5.

2022

pp. 121 - 122, 143

(1) $X \sim Y \sim N(170, 6^2)$

(a) $W = \frac{X + Y}{2}$

(b) $X \sim Y \sim W$

(2) $X \sim Y$

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \quad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$$W = XY$$

4.6.

2024

pp. 147 - 151

(1) $X \sim N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma)$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$

(2) (

)

(3)

(a) $z(0.01)$

(b) $\Phi(z_0) = 0.01 \quad z_0$

(c) $z(0.005)$

4.7.

2024

pp. 149 - 151

(1) $a \sim Z \sim N(0, 1)$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2)

X

$N(m, \sigma^2)$

$P\left(|X - m| \geq \frac{\sigma}{4}\right)$

4

(3)

m

σ

$\overline{X}$

n

$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) \leq 0.02$

n

4.8.

2024

2026

pp. 149 - 150

4

E

350,198

66.31

23.55

X

$N(66.31, (23.55)^2)$

(1)

$P(X \geq 80)$

(2)

(3-a)

(3)

80

95

(4)

20%

4.9.

2025

pp. 149 - 150

4

120

10

100

15

C

(1)

95%

C

(2)

4.10.

2022

pp. 153 - 154

(1) 6

(2) 10000

(3) 10000 4900 5100

4.11.

pp. 93 - 95, 153 - 154

200

60

6

X

(1) $42 \leq X \leq 78$

(2)

(3) X $N(60, 6^2)$ $P(42 \leq X \leq 78)$

5.

5.1.

2023 2025 pp. 168 - 169

2

5.2.

2024 pp. 93, 168 - 169

1 ATM ATM 1 40

ATM 1 60

ATM

/

60

75

90

105

120

5.3.

2023 pp. 166 - 169

(1) 3 5

(2) 1 2 1 4

5.4.

2025

4

(1) 1 3

2

$$(2) \quad X \quad P(X = 3) = 5P(X = 5)$$

$$P(X \leq 3)$$

5.5.

2023 pp. 166 - 169

1% 200

$$(1) \quad X$$

(2)

$$(3) \quad 3$$

(2) (3)

5.6.

2024 pp. 153 - 154, 166 - 169

4

$$(1) \quad 240 \quad 6 \quad 38 \quad 45$$

$$(2) \quad 0.1\% \quad 500 \quad 3$$

5.7.

2026

$$p = 0.001 \quad n = 500$$

X

$$e^{-0.5} \approx 0.6065, \quad \sqrt{0.5} \approx 0.7071$$

$$(1) \quad (a) \quad X$$

$$(b) \quad E[X] \quad \text{Var}(X)$$

(2)

(a)

$$(b) \quad P(X \geq 3)$$

(3)

(a)

(b) $P(X \geq 3)$

(4)

(a) (2) (3)

(b)

np

5.8.

2022

2023

pp. 139 - 143, 168 - 169

$p \quad 0 < p < 1$

$(1 - p)$

N

$N > 1$

(1) $N = 5$

(2) $t \quad 0 \leq t \leq N$

(3) m

$m =$

$N \quad 0 \leq m \leq N \quad m$

(4) $n \quad 0 \leq n \leq N \quad N - n \quad P(n \& N, p)$

(5) (4) n

(6) $n \quad \lambda = Np \quad N \rightarrow \infty \quad (4)$

$P(n; \lambda)$

$P(n \& N, p)$
 $\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x$

6.

6.1.

2022

p. 174

2

- (1)
- (2)

6.2.

2025

μ σ^2 n $X_1, X_2, \dots, X_n$.

(1) $X_1, X_2, \dots, X_n$

(2).

(3) n $S_n = X_1 + X_2 + \dots + X_n$.

(4)

6.3.

2025

p. 179

118020

(1) 251170

(2) N 0.91175

N

6.4.

2025

1.2 kg0.1 kg

100122.5 kg

6.5.

2022

pp. 202 - 205

μ	σ^2	3	X_1	X_2	X_3
(1)	.				
	(a)	$T_1 = X_1 + X_2 - X_3$			
	(b)	$T_2 = \frac{X_1 + 2X_2 + X_3}{4}$			
	(c)	$T_3 = \frac{X_1 + X_2 + X_3}{3}$			
	μ				
(2)	$\{T_1, T_2, T_3\}$				

6.6.

2023	2023	2025	pp. 176 - 180, 202 - 205
(1)	0	1.2	3
	$\{-1, 0, 1\}$		
	(a)		
	(b)		
	(c)		
(2)			

(3)			
(4)	θ	$\Theta_n = T(X_1, X_2, ..., X_n)$	3
	(a)		
	(b)		
	(c)		

(5)

3

6.7.

2022

2024

p. 205

1000

10

32, 86, 78, 49, 48, 77, 97, 49, 51, 74

1000

6.8.

2025

pp. 179, 216

p

n

X

(1) X .

(2) n X

(3) p_0 95%
 n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq 114)$

6.9.

2022

pp. 180 - 182

p

p

10,000

3,500

$$p = \frac{3500}{10000} = 0.35$$

(1) p

(2) p 2

6.10.

2025

- (1)

$q = \frac{Y}{n}$

A

p

A

Y

$100(1 - \alpha)\%$
- (2)

S

TV

T

140

32

T
- (a)

p

95%
- (b)

p

3%

0.95
- (c)

6.11.

2024

2024

pp. 223 - 225

- (1)

$Bin(1, p)$

$X_1, X_2, \dots, X_n$
- (a)
- (b)
- (2)

n

A

Y

$q = \frac{Y}{n}$

A

p

$100(1 - \alpha)\%$

(a)

(b)

(3)

S

TV

T

140

32

T

(a)

p

95%

(b)

p

3%

0.95

(c)
- 26 of 33

6.12.

2024

pp. 223 - 225

- (1) p 0.8 0.9 0.99
- $\hat{p}$ p 0.02
- (2) p 0.95 $\hat{p}$
- p 0.03

6.13.

2025

- 2 p S $1 - p$ F
- m S k $B(m; p)$
- (1) $B(m; p)$ $f(k)$.
- (2) X $B(m; p)$ X .
- (3) p $l(p)$.
- (4) p .
- (5) . .
- (a) (4) ?
- (b) (4) ?
- (c) (4) ?

6.14.

2025

- $N(\mu, \sigma^2)$ μ σ^2
- (1) μ
- (2) σ^2

6.15.

2025

- (1)

(2)

/

σ^2

n

N

(3)

/

7.

7.3.

2023	pp. 2235 - 236			
		8000		
	25		7700	
		5%		
		640		

7.4.

2023	pp. 238 - 240			
	T			
S	365			6
		330, 350, 385, 321, 340, 365		
	5%			

7.5.

2025	pp. 238 - 240			
	T			
S	360			6
		350, 355, 385, 390, 370, 365		

- (1)10%
- (2)5%

7.6.

2023	2024	2024	2025	pp. 235 - 236
(1)	μ	$N(\mu, 49)$	49	
	$H_0 : \mu = 4,$	$H_1 : \mu > 4$		
	m			
(a)	0.05	$m = 6$	2	

(b) 0.01 $m = 7$ 2
 (2) $H_1 : \mu = 5$ 0.05
 2 0.2 0.1

7.7.

2022 pp. 241 - 242

(1) 2 A B 2
 10 12
 A 2.04 0.03 B
 2.02 0.02 2
 5%
 2
 (2) A B
 5%

7.8.

2022 pp. 241 - 242

100 70
 68 15
 5%

7.9.

2024 pp. 252 - 253

3 1
 (1) A 10 A
 3 1 A
 5%
 (2) 10 7 3 1

7.10.

2024 pp. 252 - 253

5 1 5%

7.11.

2023

pp. 252 - 253

10 X p

$$H_0 : p = \frac{1}{2} \quad H_0$$

(1) X

(2) X

(a) $H_1 : p < \frac{1}{2}$ 5%

(b) $H_1 : p \neq \frac{1}{2}$ 2%

7.12.

2023

pp. 252 - 253

5

1 p

(1) 5 5 0

(a) $p = \frac{1}{2}$

(b) $p = \frac{2}{3}$

(2) $p \geq \frac{1}{2}$

(a) 3
 $p = \frac{1}{2}$

(b) $p = \frac{3}{4}$ 3

7.13. 2

256 116 5%

(1) 2

(2) 2 10%

7.14.

2023

2025

pp. 252 - 253

$H_0 : p = \frac{1}{2}$

$0 < p < 1$

3

H_0

3

H_0

(1) 1 1

(2) 2 2

(3) 2 $\frac{1}{3}$ p

7.15.

2024

pp. 252 - 253

3

A B C

A 3 B 2 C 1

1

1

(1) A A $\frac{1}{16}$ B B $\frac{1}{9}$ C C $\frac{1}{25}$

1

(2) 960 640

(1) 5%